ESTEBAN ANDRÉS FANTINI

Industrial Data | Cost Management & BI | ETL | IoT & Telemetry

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PROFESSIONAL SUMMARY

I have worked as a Cost Specialist in industrial environments, a role that involves working across all departments of an organization and requires strong interpersonal skills, assertive communication, and effective cross-functional coordination between the Cost Management department and other business areas. I have extensive experience in data management, processing, and reporting, having worked across different industries: Glass Manufacturing | Metalworking | Renewable Energy.

I began my professional career as an auditor in an industrial company, which allowed me to gain firsthand knowledge of its different departments and develop a comprehensive, systems-oriented understanding of the organization. This perspective later became essential in my role as Cost Coordinator, providing me with the technical skills to understand, design, and control the processes through which data is generated, helping ensure data reliability.

In recent years, I have expanded my technical knowledge into related fields, including programming, IoT, telemetry, and data engineering. I have applied these skills in various data monitoring and automation projects focused on data acquisition, transmission, validation, processing, and persistence, from its original source to remote destinations.

TECHNICAL SKILLS

- ⟶ **Industrial Data:** Process mapping, data generation points, operational KPIs, traceability, data quality, and integration between operations, data, and IT.
- ⟶ **IoT & Telemetry:** Sensors, ESP32, Raspberry Pi, MQTT, Mosquitto, Node-RED, edge devices, actuators, LoRa, and Arduino IDE.
- ⟶ **SQL:** Queries, joins, subqueries, CTEs, views.
- ⟶ **Databases:** PostgreSQL, relational data modeling, schema design, data validation, and consistency.
- ⟶ **Analytics & BI:** Advanced Excel, Tableau, exploratory data analysis, dashboards, reporting, KPIs, and variance analysis.
- ⟶ **Business & Systems:** ERP, industrial costing systems, pricing, budgeting, management control, process mapping, and business–IT integration.
- ⟶ **Data Engineering:** Python, PySpark, Databricks, ETL/ELT, data pipelines, Bronze/Silver/Gold architecture, schemas, logging, validation, and testing.
- ⟶ **Infrastructure & Tools:** Linux (Ubuntu), Bash scripting, VS Code, Notebooks, Docker, Git, GitHub.

EDUCATION

- ⟶ **University Diploma in Internet of Things (IoT)** | UTN FRRe, Argentina, 2024
- ⟶ **University Diploma in Data Science and Artificial Intelligence** | FaMAF / FCE – National University of Córdoba, 2022
- ⟶ **University Diploma in Renewable Energy** | Catholic University of Córdoba / Government of Córdoba, 2017
- ⟶ **Renewable Energy Engineering** | UNILA, Brazil, 2016 | One year completed
- ⟶ **Executive Postgraduate Program in Costing Systems Design and Management** | Escuela de Ejecutivos
- ⟶ **Accounting | Bachelor's Degree Program** | National University of Córdoba

SELECTED ADDITIONAL TRAINING

- ⟶ **Micro PLC Automation** | UTN, 2024 | Ongoing training | Signals, transducers, actuators, and PLC fundamentals.
- ⟶ **Big Data and Spark: Data Engineering with Python and PySpark** | Udemy, 2024 | Ongoing training.
- ⟶ **Artificial Intelligence Engineering** | Databricks / Indicium | Ongoing training.
- ⟶ **Apache Kafka with Python and Power BI** | Udemy, 2024 | Coursework undertaken
- ⟶ **Python, Databases, Advanced ETL & GitHub** | Udemy, 2022.
- ⟶ **CS50: Introduction to Programming with Python** | Harvard University, 2022.
- ⟶ **Multivariate Statistical Analysis** | FCE / National University of Córdoba, 2022.
- ⟶ **Introduction to Programming with C** | Online Training, 2017.
- ⟶ **Arduino** | Online Training, 2016.
- ⟶ **Import and Export Costing and Pricing** | CIRA, 2015.
- ⟶ **Computerized Systems Auditing** | FCE / National University of Córdoba, 2009.
- ⟶ **Tango and Manager II ERP Systems** | FCE / National University of Córdoba, 2008–2012.
- ⟶ **Advanced Excel** | FCE / National University of Córdoba, 2008.

LANGUAGES

Spanish: Native | **Portuguese:** Fluent | **English:** Intermediate (strong reading skills; spoken communication improving)

PROFESSIONAL EXPERIENCE

CIVE S.A. | Glass Manufacturing [Industry Leader]

Management Control, Budgeting & Cost Management | Auditing

- **I began my professional career in auditing**, reviewing internal procedures, controls, and processes across different departments, which allowed me to develop a comprehensive understanding of the organization from an early stage.
- **I audited data generation, recording, and control processes** related to purchasing, suppliers, sales, and customer accounts, using specialized audit software (ACL). I also actively supported the implementation and customization of the Manager II ERP system.
- **I subsequently specialized in Cost Management and Management Control** within a manufacturing environment involving more than 100 machines and three production lines: laminated glass, flat tempered glass, and curved tempered glass, serving the automotive, railway, and architectural sectors. My responsibilities included reporting, management dashboards, variance analysis, budgeting, make-or-buy decisions, process analysis, plant mapping, time and motion studies, and cost optimization.

Key Achievement

- **I redesigned the cost structure of a production line** following a technological upgrade and implemented an Activity-Based Costing (ABC) system.

LAMBERT HERMANOS S.A. | Metalworking Industry [4th in its sector]

Cost Management Coordinator

- **I coordinated, designed, and implemented the plant-wide Activity-Based Costing (ABC) system.**
- **I served as a liaison between IT, operations, and management**, translating business needs into data models for costing, budgeting, and management control, which were subsequently validated and integrated into the ERP system. I extracted, structured, reconciled, and validated data to develop KPIs, calculate break-even points, analyze variances, prepare budgets and operational forecasts (Budget/Forecast), optimize costs, and support operational decisions (Make-or-Buy).
- **I mapped production processes and information flows**, including time and motion studies, to identify critical activities, bottlenecks, and establish reliable methods for data collection and processing.

Key Achievements

- I redesigned cost allocation criteria, correcting distortions of up to 40% in cost distribution across products, providing the company with more accurate cost and margin information to support pricing decisions and the preparation of commercial bids.
- Through cost optimization analysis, I identified an overlooked tariff benefit related to imported materials that accounted for a significant share of total material costs, generating a positive economic and financial impact for the company. This involved Pareto analysis, auditing customs and tariff regulations, managing foreign trade administrative procedures, and operational implementation.

CASTOLDI & ASOCIADOS | Consulting

Cost Management & Business Systems Consulting

- I provide business IT solutions to clients of local accounting firms, focusing on cost management, management control, and IT solutions implementation. My work involves analyzing business processes, business rules, and data pipelines to improve operational efficiency and support business decision-making.

Key Project | E-60 Industrial Costing System

- **I developed a comprehensive costing model for a small industrial workshop manufacturing E-60 wind turbines.** The model calculates unit production costs, contribution margins, break-even points, and production budgets.
  The first version of the costing engine was developed entirely in Excel.
  I developed a second version of the system as a cost management software solution using Python, a PostgreSQL relational database, and an interactive Streamlit interface. I implemented versioned costing and cost recalculation, operational traceability, access controls, and backup and recovery procedures. (Version 1.0: validated and in production.)

SELECTED PROJECTS & RESOURCES

Personal projects through which I developed and applied skills in sensors, IoT, telemetry, device-to-device communication, and automation, aligned with Industry 4.0 environments.

TelemetriaGreenhouse | IoT + Telemetry + Data Engineering View Project

- **I designed an end-to-end greenhouse monitoring and automation system**, integrating sensors, ESP32, Raspberry Pi, MQTT, Node-RED, and remote control capabilities.
- **I subsequently enhanced the solution with a Data Engineering layer using Python and PySpark**, implementing explicit schemas, data quality validations, logging, Bronze/Silver/Gold architecture, and Parquet storage.
- **Finally, I documented the system architecture and data flows on GitHub**, using Markdown/Mermaid, UML diagrams, and the C4 model with Structurizr DSL to visually communicate the integration between IoT components and the Data Engineering layer.

Other IoT & Automation Projects

Projects focused on sensors, data acquisition, device-to-device communication, control logic, and automated actuation.

- **Automated Solar Tracker:** LDR sensors, stepper motors, and Arduino Uno for solar tracking.
- **Automated Irrigation System:** Soil moisture sensor, solenoid valve, relay module, and Arduino Uno.
- **Automatic Pet Door:** PIR sensor, electromechanical limit switches, DC motor, and Arduino Nano.
- **Emergency Communication Device:** MQTT communication and the design of a long-range LoRa network.

Analytics Dashboard | Tableau Public View Dashboard

Interactive dashboard featuring KPIs, time-series analysis, category-based analysis, and exploratory filters.